```
clc; clear; close all;

fp = 1000;
fs = 8000;
N = 50;

wc = fp/(fs/2);


b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
```

```
--- High-Pass Coefficients (Rectangular Window) ---
```

```
disp(b_rect');
```

```
   -0.0089
         0
    0.0097
    0.0143
    0.0106
         0
   -0.0117
   -0.0175
   -0.0131
         0
    0.0148
    0.0225
    0.0171
         0
   -0.0202
   -0.0314
   -0.0247
         0
    0.0318
    0.0524
    0.0445
         0
   -0.0741
   -0.1572
   -0.2223
    0.7407
   -0.2223
   -0.1572
   -0.0741
         0
    0.0445
    0.0524
    0.0318
         0
   -0.0247
   -0.0314
   -0.0202
         0
    0.0171
    0.0225
    0.0148
         0
   -0.0131
   -0.0175
```

```
   -0.0117
         0
    0.0106
    0.0143
    0.0097
         0
   -0.0089
```

```
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
```

```
--- High-Pass Coefficients (Hamming Window) ---
```

```
disp(b_hamm');
```

```
   -0.0007
         0
    0.0009
    0.0016
    0.0015
         0
   -0.0024
   -0.0044
   -0.0039
         0
    0.0060
    0.0103
    0.0088
         0
   -0.0128
   -0.0217
   -0.0184
         0
    0.0268
    0.0464
    0.0410
         0
   -0.0725
   -0.1567
   -0.2240
    0.7493
   -0.2240
   -0.1567
   -0.0725
         0
    0.0410
    0.0464
    0.0268
         0
   -0.0184
   -0.0217
   -0.0128
         0
    0.0088
    0.0103
    0.0060
         0
   -0.0039
   -0.0044
   -0.0024
```

```
         0
    0.0015
    0.0016
    0.0009
         0
   -0.0007
```

```
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
```

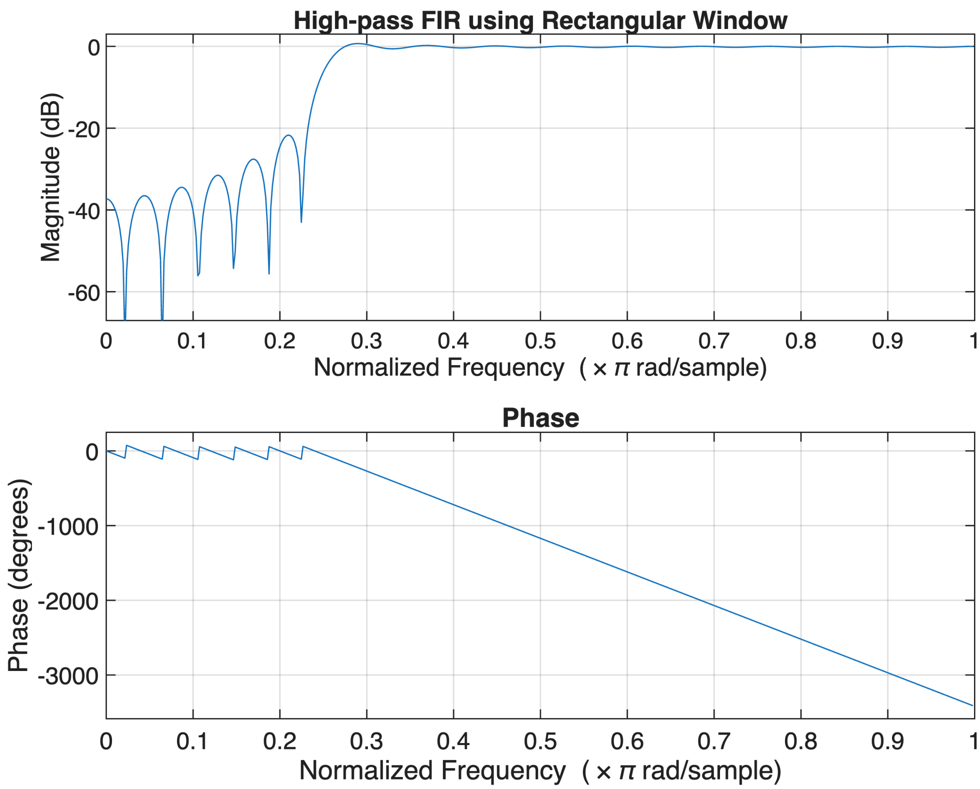
```
--- High-Pass Coefficients (Hanning Window) ---
```

```
disp(b_hann');
```

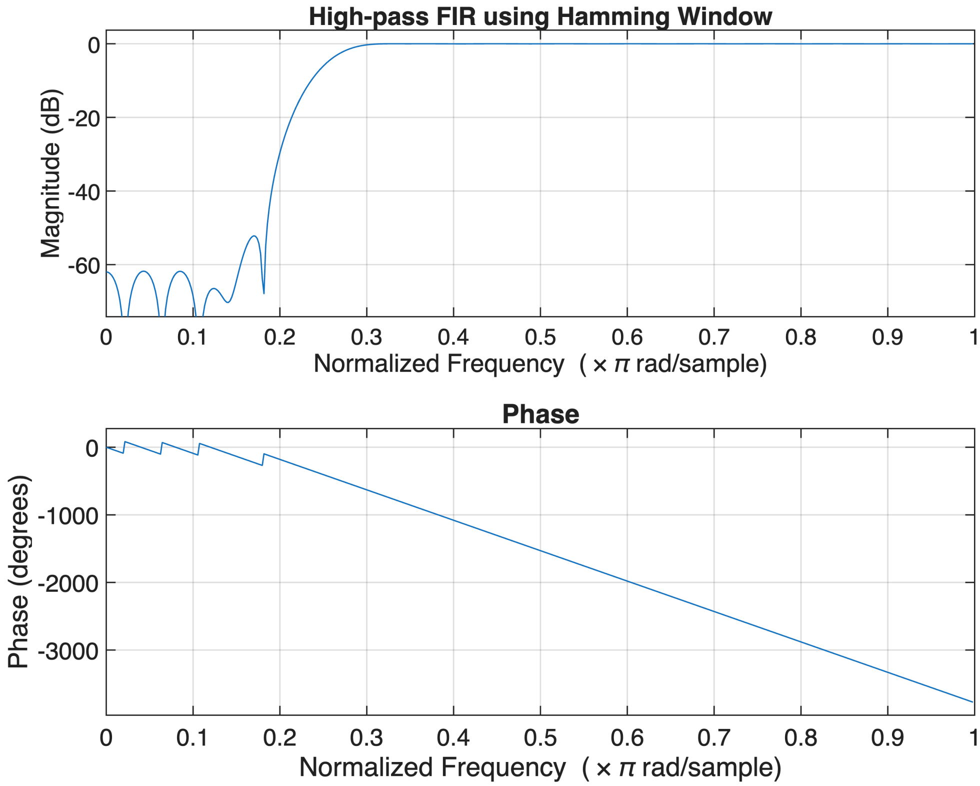
```
   -0.0000
         0
    0.0003
    0.0008
    0.0009
         0
   -0.0020
   -0.0038
   -0.0035
         0
    0.0057
    0.0100
    0.0087
         0
   -0.0127
   -0.0216
   -0.0183
         0
    0.0267
    0.0464
    0.0410
         0
   -0.0726
   -0.1568
   -0.2243
    0.7500
   -0.2243
   -0.1568
   -0.0726
         0
    0.0410
    0.0464
    0.0267
         0
   -0.0183
   -0.0216
   -0.0127
         0
    0.0087
    0.0100
    0.0057
         0
   -0.0035
   -0.0038
   -0.0020
         0
    0.0009
```

```
    0.0008
    0.0003
         0
   -0.0000
```

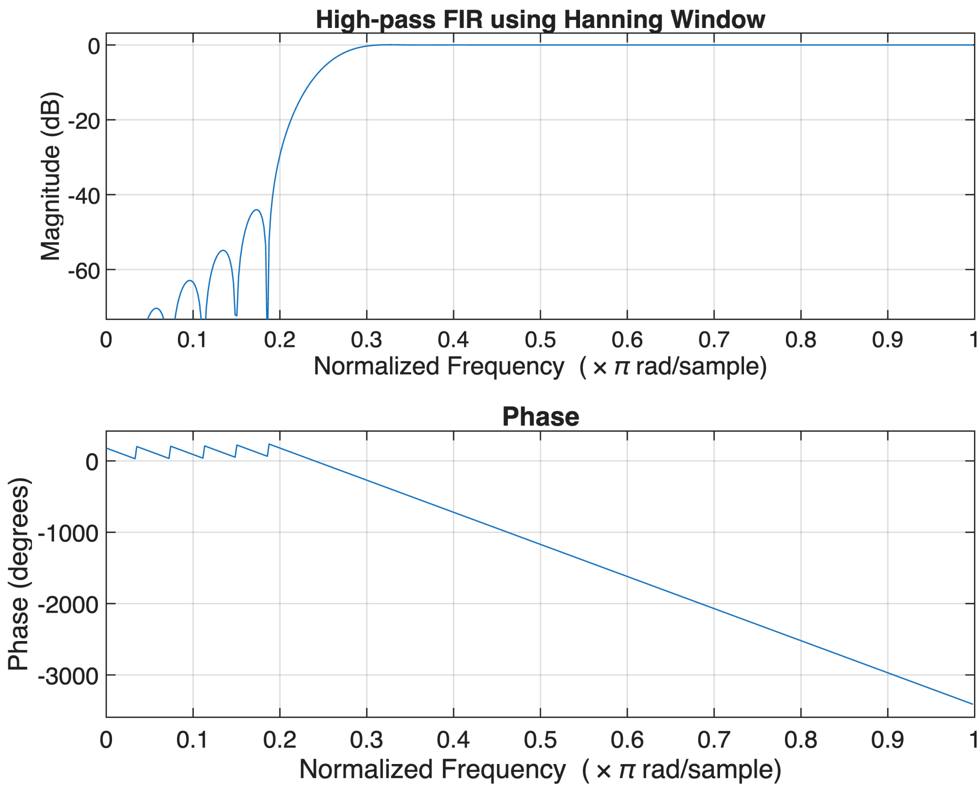
```
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
```




```
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
```




```
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```


High-pass FIR using Hanning Window
Magnitude (dB)
0
-20
-40
-60
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)